Asymptotic Upper Bound for the Predictive Risk of the Transfer Learning Estimator

1 Setup and Notation

We consider the transfer learning model described in the problem. For each site $m \in \{t, s\}$ the data satisfy

$$y^{(m)} = x^\top \beta^{(m)} + \varepsilon, \qquad x \sim \mathcal{N}(0, \Sigma), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2),$$

where $\beta^{(m)} \in \mathbb{R}^p$ is a random coefficient vector, and ε is independent of x. We observe only $x_{\mathcal{R}(t)} \in \mathbb{R}^{p_t}$ at the target site, and construct the transfer–learning estimator

$$\hat{\omega}^{(t)}_{\mathrm{TL}} = \arg\min_{\omega} \left[\frac{1}{n_t} \|Y^{(t)} - X^{(t)}_{\mathcal{R}(t)} \omega\|_2^2 + \lambda_\omega \|\omega - \hat{\omega}^{(t)}_s\|_2^2 \right],$$

with $\hat{\omega}^{(t)}_s = \Phi^{(t)} \hat{\beta}^{(s)}$ as defined in the problem. We write $n = n_t$, $p_t = |\mathcal{R}(t)|$, and adopt the shorthand

$$X = X^{(t)}_{\mathcal{R}(t)} \in \mathbb{R}^{n \times p_t}, \quad Y = Y^{(t)} \in \mathbb{R}^n, \quad \hat{\Sigma} = \tfrac{1}{n} X^\top X, \quad \lambda = \lambda_\omega, \quad \Phi = \Phi^{(t)}.$$

2 Assumptions

We collect all hypotheses used in the proof.

Assumption 2.1 (Proportional high-dimensional limit)**.** As $n \to \infty$, $p/n \to \gamma \in (0, \infty)$ and $p_t/n \to \eta \in (0, \infty)$.

Assumption 2.2 (Random regression coefficients)**.** For each $m \in \{s, t\}$,

$$\mathbb{E}[\beta^{(m)}] = 0, \qquad \mathrm{Cov}(\beta^{(m)}) = \frac{\alpha_m^2}{p} I_p, \qquad \alpha_m \neq 0,$$

and

$$\mathrm{Cov}(\beta^{(t)}, \beta^{(s)}) = \frac{\rho\, \alpha_t \alpha_s}{p} I_p, \qquad \rho \in (0, 1].$$

We assume $(\beta^{(t)}, \beta^{(s)})$ is jointly Gaussian.

Assumption 2.3 (Source estimation error)**.** $\lambda_{\max}\big(\mathbb{E}[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top]\big) \leq C^U/p$ uniformly as $(n, p) \to \infty$.

Assumption 2.4 (Independence of source and target data)**.** The source estimator $\hat{\beta}^{(s)}$ is constructed from source-site data that is independent of the target-site observations $(X, \varepsilon^{(t)})$. (Dependence through the jointly random $\beta^{(s)}, \beta^{(t)}$ is permitted.)

Assumption 2.5 (Isotropic covariance)**.** $\Sigma = I_p$. (This simplification is discussed in Remark 9.)

Assumption 2.6 (Well-separated features)**.** The index set $\mathcal{R}(t) \subseteq [p]$ is non-random, with $|\mathcal{R}(t)| = p_t$.

Assumption 2.7 (Bounded regularisation parameter)**.** $\lambda > 0$ is a deterministic constant (independent of the data) that stays bounded away from 0 and ∞ as $n \to \infty$.

3 Preliminary Lemmas

Lemma 1 (Oracle regression coefficient)**.** *Under Assumption 2.5, the best linear predictor of $y^{(t)}$ given $x_{\mathcal{R}(t)}$ (conditional on $\beta^{(t)}$) is $x_{\mathcal{R}(t)}^\top \omega^{(t)*}$ with*

$$\omega^{(t)*} = \Phi\, \beta^{(t)} = \beta^{(t)}_{\mathcal{R}(t)}.$$

The residual is $y^{(t)} - x_{\mathcal{R}(t)}^\top \omega^{(t)*} = x_{[\mathcal{R}(t)]^c}^\top \beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon$, and is independent of $x_{\mathcal{R}(t)}$.

Proof. When $\Sigma = I_p$, the blocks satisfy $\Sigma_{\mathcal{R}(t)} = I_{p_t}$, $\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c} = 0$, and the Schur complement is $\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)} = I_{p-p_t}$. Hence $\Phi = [I_{p_t} \mid 0]$, and

$$\omega^{(t)*} = \Phi\beta^{(t)} = \beta^{(t)}_{\mathcal{R}(t)}.$$

The conditional distribution $x_{[\mathcal{R}(t)]^c} \mid x_{\mathcal{R}(t)}$ has mean 0 and covariance I_{p-p_t} (by the block–diagonal structure of $\Sigma = I_p$), so $x_{[\mathcal{R}(t)]^c}$ is independent of $x_{\mathcal{R}(t)}$, and the residual $x_{[\mathcal{R}(t)]^c}^\top \beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon$ is independent of $x_{\mathcal{R}(t)}$. □

Lemma 2 (Closed form of the estimator)**.** *For any $\lambda > 0$,*

$$\hat{\omega}^{(t)}_{\mathrm{TL}} = (\hat{\Sigma} + \lambda I)^{-1}\Big(\tfrac{1}{n} X^\top Y + \lambda\, \hat{\omega}^{(t)}_s\Big).$$

Proof. The objective is strictly convex in ω. Setting the gradient to zero gives

$$-\tfrac{2}{n} X^\top (Y - X\omega) + 2\lambda(\omega - \hat{\omega}^{(t)}_s) = 0 \implies (\hat{\Sigma} + \lambda I)\omega = \tfrac{1}{n} X^\top Y + \lambda\, \hat{\omega}^{(t)}_s.$$

Since $\hat{\Sigma} + \lambda I \succ 0$, the result follows. □

Lemma 3 (Vanishing of the cross term)**.** *Under Assumptions 2.5–2.4, write $Y = X\omega^{(t)*} + \xi$ where $\xi_i = x_{i,[\mathcal{R}(t)]^c}^\top \beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon_i$. Then for every deterministic matrix M that is a function of X,*

$$\mathbb{E}\big[\xi^\top \tfrac{1}{n} X\, M\, \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) \;\big|\; X\big] = 0.$$

Proof. Since $\Sigma = I_p$, $x_{i,[\mathcal{R}(t)]^c}$ is independent of $x_{i,\mathcal{R}(t)}$, so ξ is conditionally independent of X given $(\beta^{(t)}_{[\mathcal{R}(t)]^c}, \varepsilon)$.

Write $\xi = U\beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon$ where $U \in \mathbb{R}^{n\times(p-p_t)}$ has rows $x_{i,[\mathcal{R}(t)]^c}^\top$. Since U is independent of X, of $\hat{\beta}^{(s)}$, and of ε:

$$\begin{aligned}
&\mathbb{E}[\xi^\top \tfrac{1}{n} X\, M\, \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) \mid X] \\
&\quad = \mathbb{E}\big[\beta^{(t)\top}_{[\mathcal{R}(t)]^c} \underbrace{\mathbb{E}[U^\top \mid X, \beta^{(t)}, \hat{\beta}^{(s)}]}_{=\,0} \cdot \tfrac{1}{n} X M \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) \;\big|\; X\big] \\
&\qquad + \mathbb{E}[\varepsilon^\top \tfrac{1}{n} X M \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) \mid X].
\end{aligned}$$

The first term vanishes because $\mathbb{E}[U \mid X, \beta^{(t)}, \hat{\beta}^{(s)}] = 0$ (the rows of U are mean-zero and independent of everything conditioning on). The second term vanishes because ε is independent of $(X, \hat{\beta}^{(s)}, \beta^{(t)})$. □

Lemma 4 (Bound on the transfer error covariance)**.** *Define $\delta = \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) \in \mathbb{R}^{p_t}$. Under Assumptions 2.2–2.5,*

$$\mathbb{E}[\delta\, \delta^\top] \preceq \frac{C_\delta}{p}\, I_{p_t}, \qquad C_\delta := \big(\sqrt{C^U} + \sqrt{\alpha_s^2 + \alpha_t^2 - 2\rho\, \alpha_s \alpha_t}\,\big)^2.$$

Proof. Write $\delta = \Phi(\hat{\beta}^{(s)} - \beta^{(s)}) + \Phi(\beta^{(s)} - \beta^{(t)}) =: e_{\mathcal{R}} + d_{\mathcal{R}}$.

Step 1 (Source estimation error). Since $\Phi = [I_{p_t} \mid 0]$, $e_{\mathcal{R}} = (\hat{\beta}^{(s)} - \beta^{(s)})_{\mathcal{R}(t)}$. By the interlacing inequality for principal submatrices,

$$\lambda_{\max}\big(\mathbb{E}[e_{\mathcal{R}}\, e_{\mathcal{R}}^\top]\big) \;\le\; \lambda_{\max}\big(\mathbb{E}[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top]\big) \;\le\; \frac{C^U}{p}.$$

Step 2 (Coefficient shift). $\beta^{(s)}_{\mathcal{R}(t)} - \beta^{(t)}_{\mathcal{R}(t)}$ has mean zero and covariance

$$\mathbb{E}[d_{\mathcal{R}}\, d_{\mathcal{R}}^\top] = \frac{\alpha_s^2 + \alpha_t^2 - 2\rho\,\alpha_s\alpha_t}{p}\, I_{p_t} =: \frac{\theta^2}{p}\, I_{p_t},$$

using $\mathrm{Cov}(\beta^{(s)} - \beta^{(t)}) = (\alpha_s^2/p + \alpha_t^2/p - 2\rho\alpha_s\alpha_t/p)\, I_p$.

Step 3 (Cross term bound). By the matrix Cauchy–Schwarz inequality (see, e.g., Bhatia [3]),

$$\lambda_{\max}\big(\mathbb{E}[e_{\mathcal{R}}\, d_{\mathcal{R}}^\top] + \mathbb{E}[d_{\mathcal{R}}\, e_{\mathcal{R}}^\top]\big) \;\le\; 2\,\lambda_{\max}(\mathbb{E}[e_{\mathcal{R}}\, e_{\mathcal{R}}^\top])^{1/2}\,\lambda_{\max}(\mathbb{E}[d_{\mathcal{R}}\, d_{\mathcal{R}}^\top])^{1/2} \;\le\; \frac{2\sqrt{C^U}\,\theta}{p}.$$

Step 4 (Combining). $\mathbb{E}[\delta\delta^\top] = \mathbb{E}[e_{\mathcal{R}}e_{\mathcal{R}}^\top] + \mathbb{E}[d_{\mathcal{R}}d_{\mathcal{R}}^\top] + \mathbb{E}[e_{\mathcal{R}}d_{\mathcal{R}}^\top] + \mathbb{E}[d_{\mathcal{R}}e_{\mathcal{R}}^\top]$, so

$$\lambda_{\max}\big(\mathbb{E}[\delta\delta^\top]\big) \;\le\; \frac{C^U + \theta^2 + 2\sqrt{C^U}\,\theta}{p} = \frac{(\sqrt{C^U} + \theta)^2}{p} = \frac{C_\delta}{p},$$

whence $\mathbb{E}[\delta\delta^\top] \preceq (C_\delta/p)\, I_{p_t}$. □

Lemma 5 (Marchenko–Pastur trace functionals)**.** *Suppose $X \in \mathbb{R}^{n\times p_t}$ has i.i.d. rows drawn from $\mathcal{N}(0, I_{p_t})$ and $p_t/n \to \eta \in (0,\infty)$. For fixed $\lambda > 0$, define*

$$\phi_1(\lambda,\eta) = \lim_{n\to\infty} \frac{1}{p_t}\,\mathrm{tr}\big(\hat{\Sigma}(\hat{\Sigma} + \lambda I)^{-2}\big), \qquad \phi_2(\lambda,\eta) = \lim_{n\to\infty} \frac{1}{p_t}\,\mathrm{tr}\big((\hat{\Sigma} + \lambda I)^{-2}\big).$$

Both limits exist almost surely. They are determined by the Marchenko–Pastur law F_η with density

$$dF_\eta(t) = \frac{1}{2\pi\eta t}\sqrt{(b-t)(t-a)}\;\mathbf{1}_{[a,b]}(t)\,dt + \Big(1 - \frac{1}{\eta}\Big)^+\delta_0(t), \qquad a = (1-\sqrt{\eta})^2,\; b = (1+\sqrt{\eta})^2,$$

via

$$\phi_1(\lambda,\eta) = \int_0^\infty \frac{t}{(t+\lambda)^2}\,dF_\eta(t), \qquad \phi_2(\lambda,\eta) = \int_0^\infty \frac{1}{(t+\lambda)^2}\,dF_\eta(t). \tag{1}$$

Moreover, $\phi_1 + \lambda\,\phi_2 = m_\eta(-\lambda)$, the Stieltjes transform of F_η evaluated at $-\lambda$, which satisfies the self-consistent equation

$$m_\eta(-\lambda) = \int \frac{dF_\eta(t)}{t+\lambda} = \frac{1 - \eta + \lambda - \sqrt{(1-\eta+\lambda)^2 + 4\eta\lambda}}{2\eta\lambda}, \quad \lambda > 0. \tag{2}$$

Proof. By the classical Marchenko–Pastur theorem [1], the empirical spectral distribution of $\hat{\Sigma}$ converges to F_η. The almost sure convergence of the trace functionals follows from the dominated convergence theorem applied to the a.s. converging spectral measure, together with the uniform integrability provided by the fact that $t/(t+\lambda)^2 \le 1/(4\lambda)$ and $1/(t+\lambda)^2 \le 1/\lambda^2$ are bounded continuous functions for $\lambda > 0$. The closed form (2) is classical; see [2]. □

4 Main Result

Theorem 6 (Asymptotic predictive risk upper bound)**.** *Under Assumptions 2.1–2.7, for every fixed $\lambda > 0$ the predictive risk satisfies*

$$\boxed{\limsup_{n,p\to\infty} \mathbb{E}\Big[r^{\mathrm{pred}}(\hat{\omega}_{\mathrm{TL}}^{(t)})\Big] \;\le\; \mathcal{R}_\eta\big(\lambda; C_\delta, \sigma_\xi^2\big)\,,} \tag{3}$$

where $\mathcal{R}_\eta(\lambda; a^2, \tau^2)$ denotes the Marchenko–Pastur deterministic-equivalent predictive-risk functional for isotropic ridge regression with aspect ratio η, signal strength a^2, and noise variance τ^2. Here

$$\sigma_\xi^2 = \alpha_t^2\Big(1 - \frac{\eta}{\gamma}\Big) + \sigma^2, \qquad \theta^2 = \alpha_s^2 + \alpha_t^2 - 2\rho\,\alpha_s\alpha_t, \qquad C_\delta = (\sqrt{C^U} + \theta)^2,$$

Consequently, the optimally tuned upper bound satisfies

$$\lambda^\star = \frac{\sigma_\xi^2\,\gamma}{C_\delta}. \tag{4}$$

Equivalently,

$$\boxed{\inf_{\lambda>0}\limsup_{n,p\to\infty} \mathbb{E}\Big[r^{\mathrm{pred}}(\hat{\omega}_{\mathrm{TL}}^{(t)})\Big] \;\le\; \mathcal{R}_\eta\big(\lambda^\star; C_\delta, \sigma_\xi^2\big)\,.} \tag{5}$$

5 Proof of Theorem 6

The proof proceeds in five steps.

Step 1: Bias–variance–missing feature decomposition

For a new test point $(\tilde{x}, \tilde{y}^{(t)})$ from the target site, write

$$\tilde{y}^{(t)} - \tilde{x}_{\mathcal{R}(t)}^\top \hat{\omega}_{\mathrm{TL}}^{(t)} = \underbrace{\tilde{x}_{\mathcal{R}(t)}^\top(\omega^{(t)*} - \hat{\omega}_{\mathrm{TL}}^{(t)})}_{\text{(I)}} + \underbrace{\tilde{x}_{[\mathcal{R}(t)]^c}^\top \beta_{[\mathcal{R}(t)]^c}^{(t)}}_{\text{(II)}} + \underbrace{\tilde{\varepsilon}}_{\text{(III)}}.$$

By Lemma 1, (II) is independent of $\tilde{x}_{\mathcal{R}(t)}$, and (III) is independent of everything else. Squaring and taking $\mathbb{E}[\cdot \mid X]$:

- The cross term between (I) and (III) vanishes because $\mathbb{E}[\tilde{\varepsilon}] = 0$ and $\tilde{\varepsilon}$ is independent of everything else.
- The cross term between (I) and (II): (II) depends on $\tilde{x}_{[\mathcal{R}(t)]^c}$ and $\beta_{[\mathcal{R}(t)]^c}^{(t)}$; (I) depends on $\tilde{x}_{\mathcal{R}(t)}$ and the training data. Conditioning first on all randomness except $\tilde{x}_{[\mathcal{R}(t)]^c}$, we get $\mathbb{E}[\tilde{x}_{[\mathcal{R}(t)]^c} \mid \tilde{x}_{\mathcal{R}(t)}] = 0$ (by $\Sigma = I_p$), so this cross term vanishes.
- The cross term between (II) and (III) vanishes since $\tilde{\varepsilon} \perp \tilde{x}$.

Therefore

$$r^{\mathrm{pred}}(\hat{\omega}_{\mathrm{TL}}^{(t)}) = \underbrace{\mathbb{E}\big[(\omega^{(t)*} - \hat{\omega}_{\mathrm{TL}}^{(t)})^\top \Sigma_{\mathcal{R}(t)} (\omega^{(t)*} - \hat{\omega}_{\mathrm{TL}}^{(t)}) \;\big|\; X\big]}_{R_{\mathrm{est}}} + \underbrace{\mathbb{E}\big[\beta_{[\mathcal{R}(t)]^c}^{(t)\top} \Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}\, \beta_{[\mathcal{R}(t)]^c}^{(t)}\big]}_{R_{\mathrm{miss}}} + \sigma^2, \tag{6}$$

where we used $\mathbb{E}[\tilde{x}_{\mathcal{R}(t)}\tilde{x}_{\mathcal{R}(t)}^\top] = \Sigma_{\mathcal{R}(t)} = I_{p_t}$ and $\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)} = I_{p-p_t}$.

The missing-feature risk is

$$R_{\mathrm{miss}} = \mathrm{tr}\big(I_{p-p_t} \cdot \tfrac{\alpha_t^2}{p}\, I_{p-p_t}\big) = \frac{\alpha_t^2(p-p_t)}{p} = \alpha_t^2\Big(1 - \frac{p_t}{p}\Big) \xrightarrow{n\to\infty} \alpha_t^2\Big(1 - \frac{\eta}{\gamma}\Big).$$

Step 2: Estimation error decomposition

By Lemma 2 and the identity $Y = X\omega^{(t)*} + \xi$ (Lemma 1), set $M = (\hat{\Sigma} + \lambda I)^{-1}$ and compute

$$\begin{aligned}\hat{\omega}_{\mathrm{TL}}^{(t)} - \omega^{(t)*} &= M\big(\hat{\Sigma}\,\omega^{(t)*} + \tfrac{1}{n}X^\top\xi + \lambda\,\hat{\omega}_s^{(t)}\big) - \omega^{(t)*} \\ &= M\big(\tfrac{1}{n}X^\top\xi + \lambda(\hat{\omega}_s^{(t)} - \omega^{(t)*})\big) \\ &= \underbrace{M\,\tfrac{1}{n}X^\top\xi}_{A} \;+\; \underbrace{\lambda\,M\,\delta}_{B},\end{aligned} \tag{7}$$

where $\delta = \Phi(\hat{\beta}^{(s)} - \beta^{(t)}) = \hat{\beta}^{(s)}_{\mathcal{R}(t)} - \beta^{(t)}_{\mathcal{R}(t)}$.

Since $\Sigma_{\mathcal{R}(t)} = I_{p_t}$,

$$R_{\mathrm{est}} = \mathbb{E}[\|A + B\|^2 \mid X] = \mathbb{E}[\|A\|^2 \mid X] + \mathbb{E}[\|B\|^2 \mid X] + 2\,\mathbb{E}[A^\top B \mid X].$$

By Lemma 3, the cross term $\mathbb{E}[A^\top B \mid X] = 0$. Hence

$$R_{\mathrm{est}} = \underbrace{\mathbb{E}[\|A\|^2 \mid X]}_{T_1(\text{variance})} + \underbrace{\mathbb{E}[\|B\|^2 \mid X]}_{T_2(\text{bias})}. \tag{8}$$

Step 3: Evaluation of the variance term T_1

We compute $\mathbb{E}[\xi\xi^\top \mid X]$. Since $\Sigma = I_p$, the rows of U (the matrix with rows $x_{i,[\mathcal{R}(t)]^c}^\top$) are i.i.d. $\mathcal{N}(0, I_{p-p_t})$ and *independent* of X. With ε also independent:

$$\mathbb{E}[\xi_i\xi_j \mid X] = \begin{cases}\mathbb{E}\big[(x_{i,[\mathcal{R}(t)]^c}^\top\beta^{(t)}_{[\mathcal{R}(t)]^c})^2\big] + \sigma^2 & \text{if } i = j, \\ 0 & \text{if } i \neq j,\end{cases}$$

since $\mathbb{E}[x_{i,[\mathcal{R}(t)]^c}] = 0$ independently across i, and $\varepsilon_i \perp \varepsilon_j$ for $i \neq j$.

For the diagonal entry:

$$\mathbb{E}\big[(x_{i,[\mathcal{R}(t)]^c}^\top\beta^{(t)}_{[\mathcal{R}(t)]^c})^2\big] = \mathbb{E}\big[\beta^{(t)\top}_{[\mathcal{R}(t)]^c}\underbrace{\mathbb{E}[x_{i,[\mathcal{R}(t)]^c}x_{i,[\mathcal{R}(t)]^c}^\top]}_{I_{p-p_t}}\beta^{(t)}_{[\mathcal{R}(t)]^c}\big] = \mathbb{E}[\|\beta^{(t)}_{[\mathcal{R}(t)]^c}\|^2] = \frac{\alpha_t^2(p-p_t)}{p}.$$

Define $\sigma_\xi^2 := \alpha_t^2(p-p_t)/p + \sigma^2$. Then $\mathbb{E}[\xi\xi^\top \mid X] = \sigma_\xi^2\,I_n$, and

$$\begin{aligned}T_1 &= \mathbb{E}\big[\tfrac{1}{n^2}\xi^\top X M^2 X^\top\xi \;\big|\; X\big] = \frac{\sigma_\xi^2}{n^2}\operatorname{tr}(XM^2X^\top) = \frac{\sigma_\xi^2}{n}\operatorname{tr}(M^2\hat{\Sigma}) \\ &= \frac{\sigma_\xi^2}{n}\operatorname{tr}\big(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-2}\big) = \sigma_\xi^2\,\frac{p_t}{n}\cdot\frac{1}{p_t}\operatorname{tr}\big(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-2}\big).\end{aligned} \tag{9}$$

By Lemma 5, $\frac{1}{p_t}\operatorname{tr}(\hat{\Sigma}(\hat{\Sigma}+\lambda I)^{-2}) \xrightarrow{a.s.} \phi_1(\lambda,\eta)$, so

$$T_1 \;\xrightarrow{a.s.}\; \sigma_\xi^2\,\eta\,\phi_1(\lambda,\eta). \tag{10}$$

Step 4: Upper bound on the bias term T_2

$$T_2 = \lambda^2\,\mathbb{E}[\delta^\top M^2\delta \mid X] = \lambda^2\,\operatorname{tr}\big(M^2\,\mathbb{E}[\delta\delta^\top]\big). \tag{11}$$

(Here M is a deterministic function of X, and δ is independent of X only through the residual randomness of β and $\hat{\beta}^{(s)}$. The expectation is over all remaining randomness.)

By Lemma 4, $\mathbb{E}[\delta\delta^\top] \preceq (C_\delta/p)\, I_{p_t}$, hence

$$T_2 \le \frac{\lambda^2 C_\delta}{p}\ \mathrm{tr}\big((\hat{\Sigma}+\lambda I)^{-2}\big) = \frac{\lambda^2 C_\delta\, p_t}{p}\cdot\frac{1}{p_t}\,\mathrm{tr}\big((\hat{\Sigma}+\lambda I)^{-2}\big). \tag{12}$$

By Lemma 5, $\frac{1}{p_t}\,\mathrm{tr}((\hat{\Sigma}+\lambda I)^{-2}) \xrightarrow{a.s.} \phi_2(\lambda,\eta)$, and $p_t/p \to \eta/\gamma$, so

$$T_2 \ \le\ \frac{\lambda^2 C_\delta\,\eta}{\gamma}\,\phi_2(\lambda,\eta) + o(1). \tag{13}$$

Step 5: Combining and optimising

Substituting (10) and (13) into (6)–(8), and noting that $\sigma_\xi^2 \to \alpha_t^2(1-\eta/\gamma)+\sigma^2$, we obtain the claimed bound (3).

Optimisation over λ. Define the estimation-error upper bound

$$\overline{R}_{\mathrm{est}}(\lambda) = \sigma_\xi^2\,\eta\,\phi_1(\lambda,\eta) + \frac{\lambda^2 C_\delta\,\eta}{\gamma}\,\phi_2(\lambda,\eta).$$

Setting $\partial\overline{R}_{\mathrm{est}}/\partial\lambda = 0$:

$$\sigma_\xi^2\,\eta\,\phi_1'(\lambda,\eta) + \frac{C_\delta\,\eta}{\gamma}\big(2\lambda\,\phi_2(\lambda,\eta) + \lambda^2\phi_2'(\lambda,\eta)\big) = 0. \tag{14}$$

The identity $\phi_1(\lambda,\eta) = \int t(t+\lambda)^{-2}\,dF_\eta$ yields $\phi_1'(\lambda,\eta) = -2\int t(t+\lambda)^{-3}\,dF_\eta$, and similarly $\phi_2'(\lambda,\eta) = -2\int(t+\lambda)^{-3}\,dF_\eta$. Hence $\phi_1' = -2\int t(t+\lambda)^{-3}$ and $2\phi_2 + \lambda\phi_2' = 2\int(t+\lambda)^{-2} - 2\lambda\int(t+\lambda)^{-3} = 2\int t(t+\lambda)^{-3} = -\phi_1'$.

Substituting into (14):

$$\sigma_\xi^2\,\eta\,\phi_1' + \frac{C_\delta\,\eta\,\lambda}{\gamma}(-\phi_1') = 0 \ \ \Longrightarrow\ \ \phi_1'\Big(\sigma_\xi^2 - \frac{C_\delta\,\lambda}{\gamma}\Big) = 0.$$

Since $\phi_1'(\lambda,\eta) < 0$ for all $\lambda > 0$ (the integrand is strictly positive on the support of F_η), we conclude

$$\boxed{\lambda^\star = \frac{\sigma_\xi^2\,\gamma}{C_\delta}.} \tag{15}$$

Optimal risk. Substituting $\lambda^\star$ into $\overline{R}_{\mathrm{est}}$:

$$\begin{aligned}\overline{R}_{\mathrm{est}}(\lambda^\star) &= \sigma_\xi^2\,\eta\,\phi_1(\lambda^\star,\eta) + \frac{(\sigma_\xi^2\gamma/C_\delta)^2\,C_\delta\,\eta}{\gamma}\,\phi_2(\lambda^\star,\eta)\\ &= \sigma_\xi^2\,\eta\big[\phi_1(\lambda^\star,\eta) + \lambda^\star\,\phi_2(\lambda^\star,\eta)\big]\\ &= \sigma_\xi^2\,\eta\,m_\eta(-\lambda^\star),\end{aligned} \tag{16}$$

where the last equality uses $\phi_1 + \lambda\phi_2 = \int \frac{t+\lambda}{(t+\lambda)^2}\,dF_\eta = \int \frac{1}{t+\lambda}\,dF_\eta = m_\eta(-\lambda)$.

Thus the optimally tuned predictive-risk upper bound can be reported in the unexpanded deterministic-equivalent form

$$\boxed{\inf_{\lambda>0}\ \limsup_{n,p\to\infty}\mathbb{E}\Big[r^{\mathrm{pred}}(\hat{\omega}_{\mathrm{TL}}^{(t)})\Big]\ \le\ \mathcal{R}_\eta\big(\lambda^\star; C_\delta, \sigma_\xi^2\big)\,, \qquad \lambda^\star = \frac{\sigma_\xi^2\,\gamma}{C_\delta}.} \tag{17}$$

This completes the proof.

6 Remarks

Remark 7 (Interpretation of $\lambda^\star$)**.** *The optimal regularisation $\lambda^\star = \sigma_\xi^2\gamma/C_\delta$ balances the effective noise level σ_ξ^2 (which includes the irreducible error from missing features) against the transfer error C_δ. When the source estimator is very accurate ($C^U \to 0$) and the sites are nearly identical ($\rho \to 1$, $\alpha_s = \alpha_t$), $C_\delta \to 0$, so $\lambda^\star \to \infty$: we rely almost entirely on the transferred estimate. Conversely, when the source information is poor (C_δ large), $\lambda^\star \to 0$ and the estimator reduces to the ordinary least squares (or Moore–Penrose) solution on the target data alone.*

Remark 8 (Comparison with no transfer)**.** *Without transfer ($\lambda = 0$, provided $\eta < 1$ so that $\hat{\Sigma}$ is invertible), the estimation error is $\sigma_\xi^2\eta\phi_1(0,\eta)$. Since $\phi_1(0,\eta) = \int t^{-1}\, dF_\eta(t)$, which diverges as $\eta \nearrow 1$, the transfer estimator with optimal $\lambda^\star > 0$ provides a strict improvement whenever η is not too small and C_δ is finite. When $\eta \geq 1$, the no-transfer estimator is not even defined (the OLS problem is underdetermined), while the transfer estimator remains well-defined for any $\lambda > 0$.*

Remark 9 (General covariance Σ)**.** *When $\Sigma \neq I_p$, the analysis extends with the following modifications: (i) the oracle coefficient becomes $\omega^{(t)*} = \beta^{(t)}_{\mathcal{R}(t)} + \Sigma^{-1}_{\mathcal{R}(t)}\Sigma_{\mathcal{R}(t),[\mathcal{R}(t)]^c}\,\beta^{(t)}_{[\mathcal{R}(t)]^c} = \Phi\beta^{(t)}$; (ii) the residual covariance becomes $\sigma_\xi^2 = (\alpha_t^2/p)\,\mathrm{tr}(\Sigma_{[\mathcal{R}(t)]^c|\mathcal{R}(t)}) + \sigma^2$; (iii) the trace functionals ϕ_1, ϕ_2 are replaced by their generalisations for the generalised Marchenko–Pastur law associated with $\Sigma_{\mathcal{R}(t)}$; and (iv) the transfer error bound requires $\mathbb{E}[\delta\delta^\top] \preceq (C_\delta/p)\,\Phi\Phi^\top$, which involves $\|\Phi\|_{\mathrm{op}}$ and the condition number of $\Sigma_{\mathcal{R}(t)}$. The structure of the proof is identical; the additional spectral assumptions on Σ (bounded eigenvalues, bounded away from 0) ensure the random-matrix-theory limits exist.*

Remark 10 (Sharpness of the bound)**.** *The bound is an* equality *(up to the $o(1)$ term) under two conditions: (a) the cross terms in Lemma 4 vanish, i.e. $\mathbb{E}[e_{\mathcal{R}}\, d_{\mathcal{R}}^\top] = 0$ (which holds, for example, when $\hat{\beta}^{(s)}$ is an unbiased estimator of $\beta^{(s)}$ constructed independently of the random coefficients), and (b) $\mathbb{E}[\delta\delta^\top] = (C_\delta/p)\, I_{p_t}$, which holds when $\mathbb{E}[e_{\mathcal{R}} e_{\mathcal{R}}^\top] = (C^U/p)\, I_{p_t}$ exactly (e.g. when the source estimator is a ridge estimator with isotropic error covariance). In that case the bound* (17) *is the exact asymptotic risk.*

References

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